微生物检测文献精读

VITEK MITUBE 与 MALDI-TOF 阳性血培养液的细菌鉴定

BL0060 · 装置、信号、分类与临床验证

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在线发表 2025-07-18 · 卷期 2025年8月 · DOI 10.1128/jcm.00122-25

Rhoads DD et al. J Clin Microbiol. 2025;63(8):e00122-25. DOI: 10.1128/jcm.00122-25
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内容

检测任务与装置

03–10　阳性培养液输入、样本清理与生物量门槛

质谱与分类

11–16　MALDI、飞行时间、加权分箱与拒识

研究设计与结果

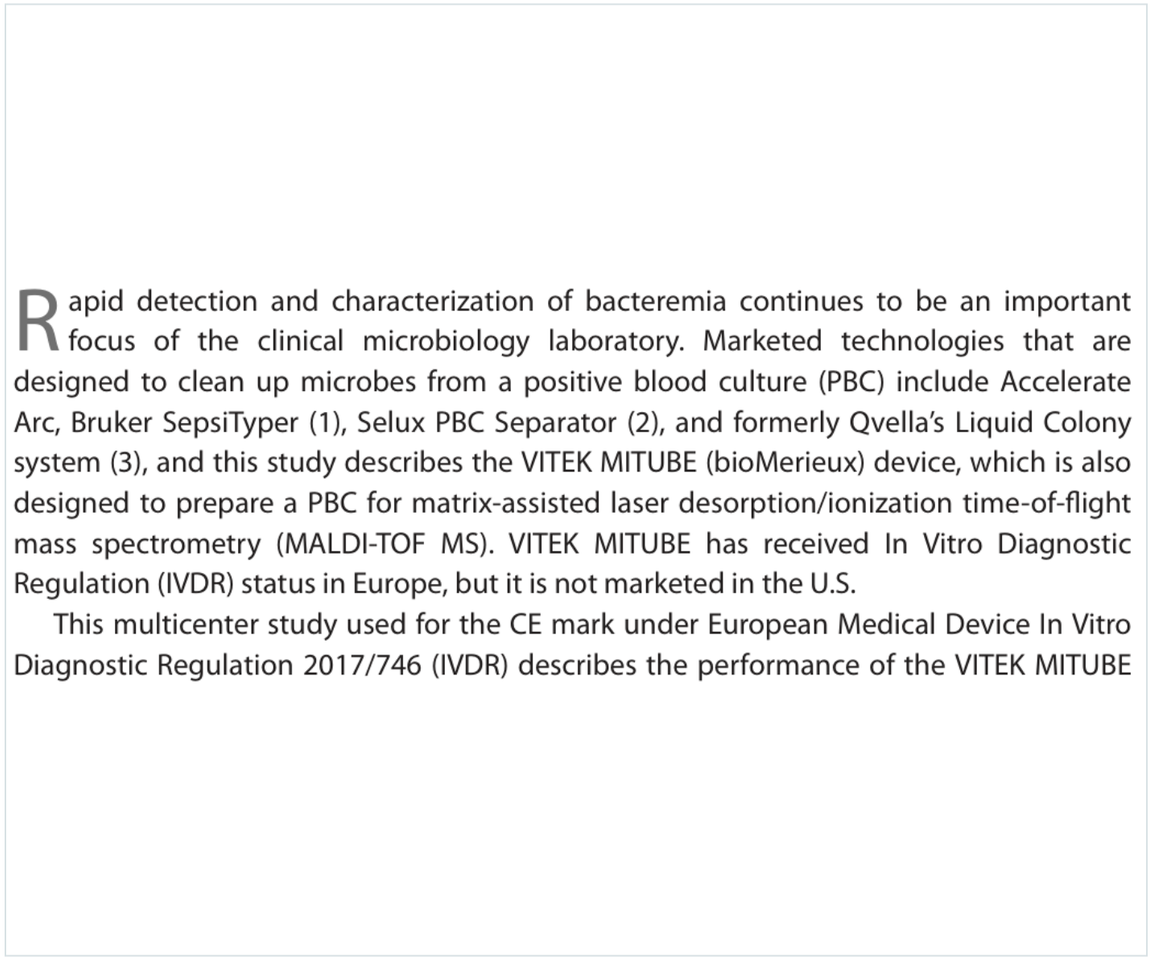
17–28　适用菌种、参照方法、样本分母及失败结果

应用边界与研究支持

29–33　证据限制、introduction 定位及来源

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检测任务：缩短培养报阳后的鉴定环节



Rapid detection and characterization of bacteremia continues to be an important focus of the clinical microbiology laboratory. Marketed technologies that are designed to clean up microbes from a positive blood culture (PBC) include Accelerate Arc, Bruker SepsiTyper (1), Selux PBC Separator (2), and formerly Qvella's Liquid Colony system (3), and this study describes the VITEK MITUBE (bioMerieux) device, which is also designed to prepare a PBC for matrix-assisted laser desorption/ionization time-of-flight mass spectrometry (MALDI-TOF MS). VITEK MITUBE has received In Vitro Diagnostic Regulation (IVDR) status in Europe, but it is not marketed in the U.S.

This multicenter study used for the CE mark under European Medical Device In Vitro Diagnostic Regulation 2017/746 (IVDR) describes the performance of the VITEK MITUBE

原文截图 · PDF p.1，Introduction

输入已经培养报阳

论文面对的是阳性血培养液（PBC），不是直接检测未经培养的原始血液。培养阶段已经产生了可检出的微生物生物量。

需要改变的环节

常规参照流程先获得平板上的分离菌落，再做质谱鉴定。MITUBE 将培养液清理成适合质谱的细菌悬液，研究其能否省去鉴定前的再培养。

本文主要评价菌种鉴定结果；未测量患者结局，也未证明从采血到报告的总时间缩短多少。

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Fig.1：检测链与仪器接口

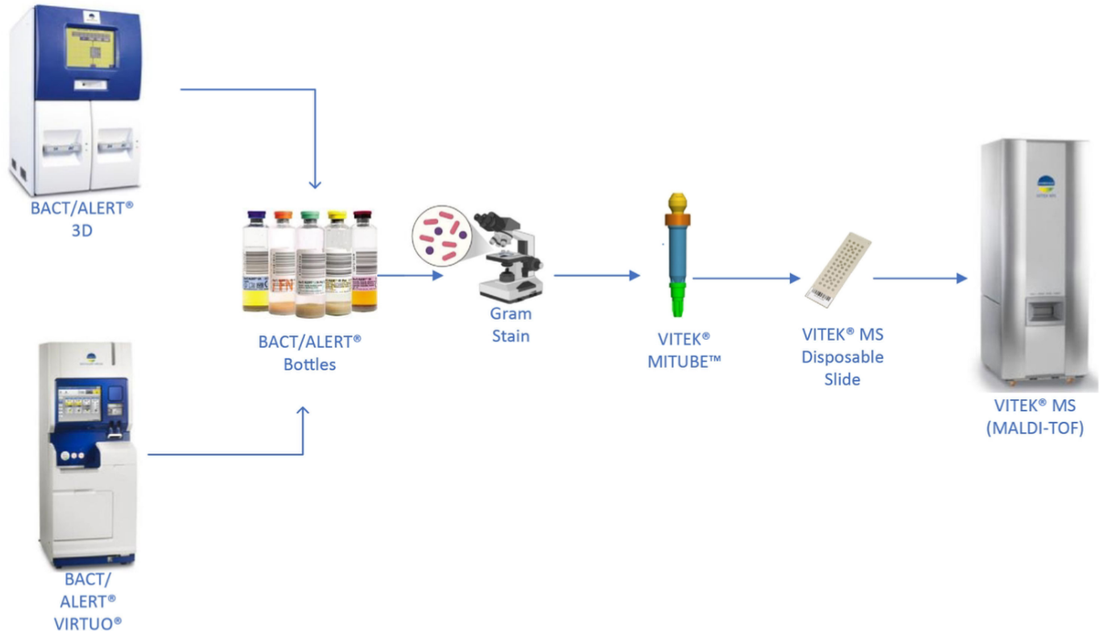


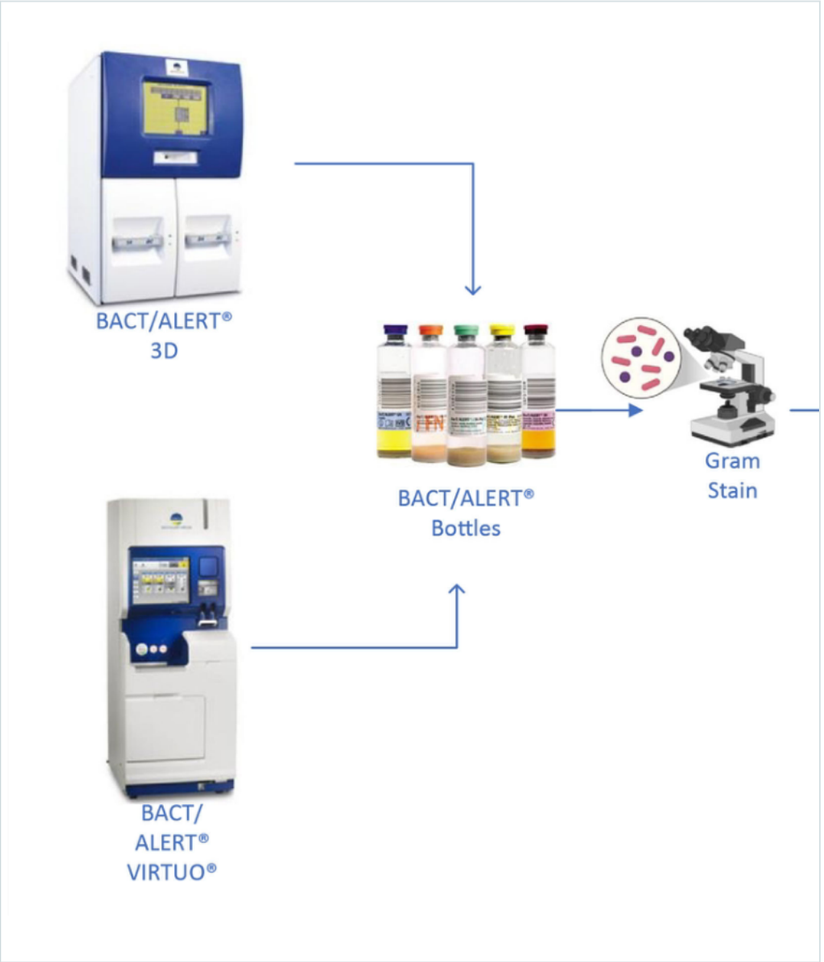
FIG 1 Sample flow for VITEK MITUBE testing in the prospective clinical study. The sample workflow from positive blood culture bottle to MALDI-TOF mass spectrometric identification is depicted. Once a blood culture bottle signals positive by an automated blood culture monitoring system, then the positive blood culture broth is removed. A Gram stain is performed, and cultures that appear to be monomicrobial gram-negative bacilli are carried forward to VITEK MITUBE for microbial clean-up, which results in a saline suspension of bacteria (see Fig. 2 for more details). The suspension is used to inoculate a MALDI target, which is then tested to identify the gram-negative bacteria.

原文截图 · PDF p.2，Fig.1 完整图与图注，无字母子图

培养仪报阳 → 革兰染色筛选 → MITUBE清理 → MALDI靶板 → VITEK MS鉴定。图中箭头表示样本流转，未给出端到端时间。

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Fig.1前段：培养报阳与输入筛选



原文截图 · PDF p.2，Fig.1前段区域放大（非原文字母子图）

FIG 1 Sample flow for VITEK MITUBE testing in the prospective clinical study. The sample workflow from positive blood culture bottle to MALDI-TOF mass spectrometric identification is depicted. Once a blood culture bottle signals positive by an automated blood culture monitoring system, then the positive blood culture broth is removed. A Gram stain is performed, and cultures that appear to be monomicrobial gram-negative bacilli are carried forward to VITEK MITUBE for microbial clean-up, which results in a saline suspension of bacteria (see Fig. 2 for more details). The suspension is used to inoculate a MALDI target, which is then tested to identify the gram-negative bacteria.

原文截图 · PDF p.2，Fig.1 完整图注

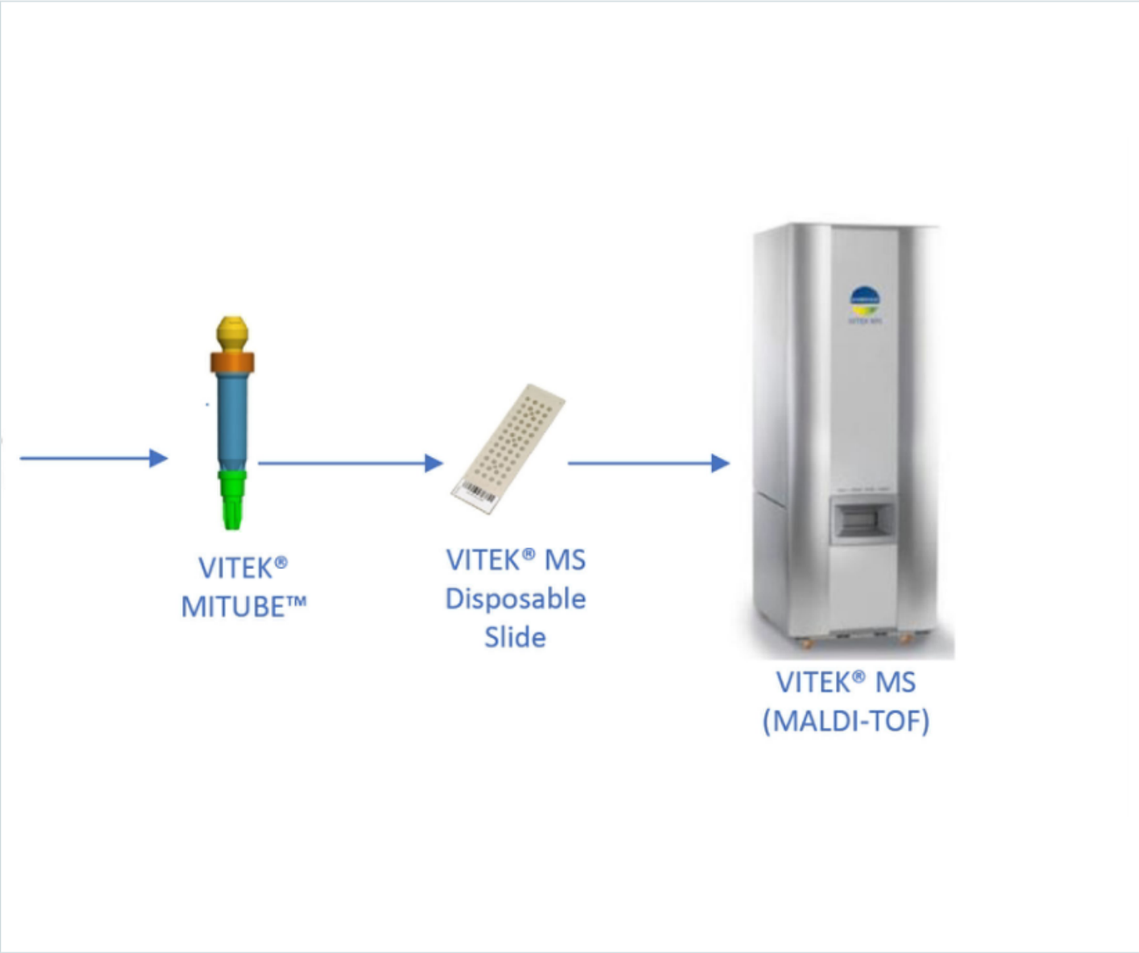
培养与形态筛选

图左两种培养仪都先监测培养瓶是否报阳。革兰染色用于判断是否符合单一革兰阴性杆菌的输入条件。

这些是待测路径的入口条件；前段的培养与筛选成本不应从完整工作流中消失。

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Fig.1后段：从悬液到质谱输出



原文截图 · PDF p.2，Fig.1后段区域放大（非原文字母子图）

装置与读出分工

MITUBE输出经过清理的细菌悬液；一次性靶板是样本与质谱仪的接口。VITEK MS产生谱并执行知识库分类。

清理装置、MALDI靶点与分类软件是不同模块；任一模块失败都可能使整个检测链没有可靠结果。

放大图只显示后段；上一完整定位页保留全部图形与箭头。

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Fig.1图注与检测时间的起止点

FIG 1 Sample flow for VITEK MITUBE testing in the prospective clinical study. The sample workflow from positive blood culture bottle to MALDI-TOF mass spectrometric identification is depicted. Once a blood culture bottle signals positive by an automated blood culture monitoring system, then the positive blood culture broth is removed. A Gram stain is performed, and cultures that appear to be monomicrobial gram-negative bacilli are carried forward to VITEK MITUBE for microbial clean-up, which results in a saline suspension of bacteria (see Fig. 2 for more details). The suspension is used to inoculate a MALDI target, which is then tested to identify the gram-negative bacteria.

原文截图 · PDF p.2，Fig.1 完整图注

图注定义的输入与输出

起点是自动血培养仪已经报阳的培养瓶；终点是质谱给出菌种鉴定。中间包含染色筛选、微生物清理和靶板输入。

因此，本研究的“direct”指从阳性培养液进入鉴定，不表示省去了此前的培养。

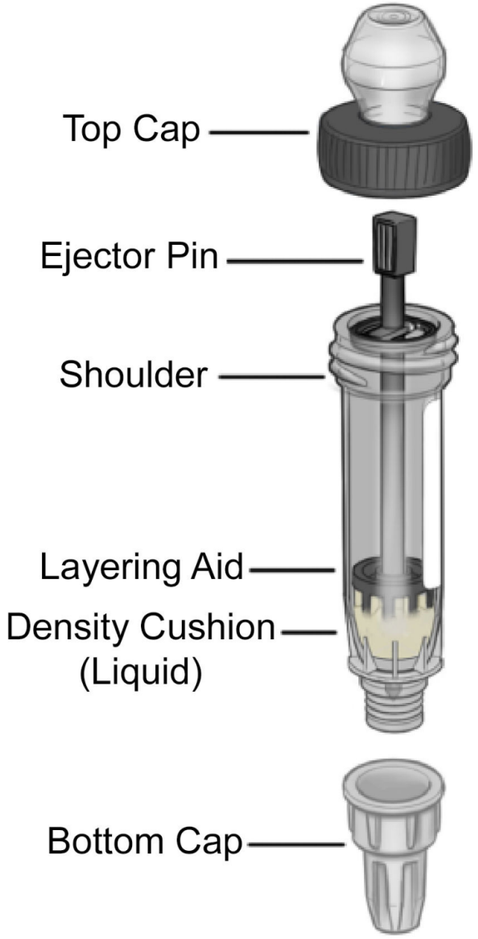
时间指标需另行验证

论文没有报告从报阳到最终结果的逐样本时间分布，也没有统计从采血起算的总诊断时间。

No ID之后的重复采集可能增加完成时间；首次采集成功率与最终识别率应与各自时间边界一同记录。

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Fig.2：MITUBE的完整结构与图注



原文截图 · PDF p.4，Fig.2完整装置图（完整图注见右侧）

FIG 2 Exploded diagram of the VITEK MITUBE device. The components of the VITEK MITUBE device are labeled. The specimen processing workflow is as follows. After removing the top cap, the device is loaded with 6 mL of positive blood culture broth containing gram-negative bacteria that has been mixed 1:2 with lysis buffer. Then, the top lid is returned to the device, and it is centrifuged for 10 minutes. After centrifugation, the bottom cap is removed, the ejector pin is twisted, and the microbial pellet is dispensed into 1 mL of saline. Then, 1.0 µL of bacterial saline suspension is added to a pre-warmed VITEK MS target, followed by formic acid and then matrix. The bacterial isolate is then identified using VITEK MS.

原文截图 · PDF p.4，Fig.2 完整图注

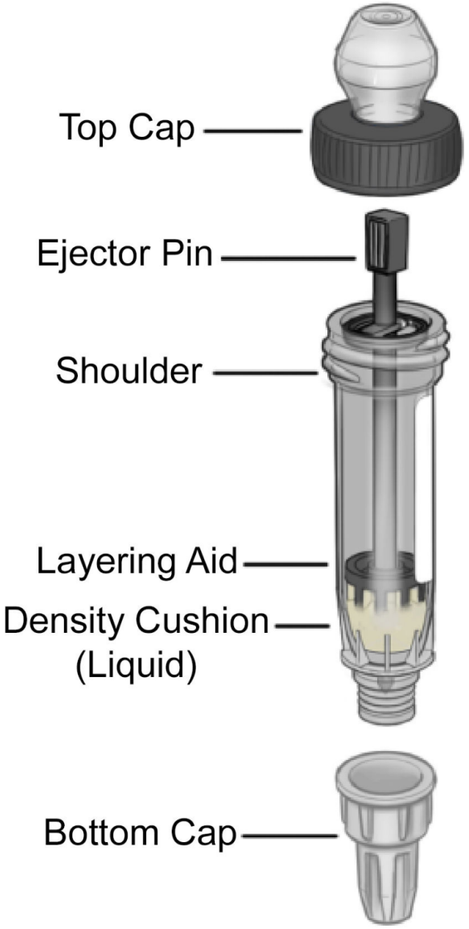
读图定位

顶盖封闭装载端；顶针贯穿主体，用于释放底部沉淀。肩部、分层辅助件、液体密度垫与底盖共同构成样本分离及回收接口。

此页保留整幅装置图和完整图注。中文讲解说明部件作用，不构成操作规程。

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微生物分离与浓集的作用



原文截图 · PDF p.4，Fig.2 完整装置图（图注另页放大）

样本清理

输入含培养液、血液来源成分和细菌。经裂解液处理及密度垫离心分离，装置回收微生物沉淀，再形成细菌悬液。

与质谱信号的关系

需要让菌体相关信号在靶点中足够强，并减少培养基及宿主背景干扰。这是对装置用途的工程解释；论文没有量化宿主去除率、细菌回收率或浓缩倍数。

顶针与下端出口将分离后的沉淀转移到下一检测模块。装置本身不输出菌种名称。

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生物量门槛决定哪些样本进入鉴定

connected, was shaken to rinse the microbial pellet from the ejector pin. The saline tube and VITEK MITUBE were separated, and the saline tube was capped and vortexed for 3 seconds. Biomass in the saline tube was confirmed to be adequate if ≥0.50 McFarland was measured using the VITEK DENSICHEK. If the <0.50 McFarland was identified, then the sample was considered to have “low biomass” that was inadequate for testing. If the sample had adequate biomass, then a VITEK MS Slide was then prepared by spotting two different spots with 1.0 µL of suspension onto a MALDI target. The same spots received 0.5 µL of formic acid (VITEK MS-FA reagent, bioMérieux) and also 1.0 µL of VITEK MS-CHCA matrix (bioMérieux). The slides were then tested using the VITEK MS or VITEK MS PRIME instrument. A subset of the existing MALDI knowledge base was used, which included a limited number of taxa (Box 1).

原文截图 · PDF p.3，Materials and Methods：生物量与质谱输入

论文使用的门槛

悬液经 DENSICHEK 测量达到 ⩾0.50 McFarland 才进入后续检测。低于门槛被归为低生物量，不计入最终125例分析集。

量的含义与缺失

McFarland 是浊度标准，不是本研究测得的 CFU/mL 检出限。不能把0.50换算为本文的LOD。

147例候选中有11例因低生物量排除；标本进入质谱前已经存在一次选择。

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MALDI：基质辅助解吸与离子化

sample had adequate biomass, then a VITEK MS Slide was then prepared by spotting two different spots with 1.0 µL of suspension onto a MALDI target. The same spots received 0.5 µL of formic acid (VITEK MS-FA reagent, bioMérieux) and also 1.0 µL of VITEK MS-CHCA matrix (bioMérieux). The slides were then tested using the VITEK MS or VITEK MS PRIME instrument. A subset of the existing MALDI knowledge base was used, which included a limited number of taxa (Box 1).

原文截图 · PDF p.3，Materials and Methods：MALDI靶点与知识库

输入与能量转换

靶点包含菌体相关分子与CHCA基质。激光脉冲的能量由基质吸收，使样本相关分子进入气相，并通过离子形成过程得到可加速的带电分子。

输出不是细菌图像

检测器接收的是离子；后续得到强度随质荷比变化的谱。激光用于产生离子，本文读出属于质谱，而非通过显微图像识别细菌。

论文没有给出激光波长、脉冲能量或原始质谱；不能从Fig.1推导这些参数。

基础物理：Shimadzu，MALDI Theory and Basics，I. MALDI Theory；2026-10-27查阅。https://www.ssi.shimadzu.com/service-support/faq/maldi/maldi-basics/index.html

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TOF：飞行时间如何编码质荷比

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原文截图 · PDF p.3，Materials and Methods：MALDI靶点与知识库

理想化物理关系

$qU = \frac{1}{2}mv^2$，$t = L/v$

$t = L\sqrt{[m/(2qU)]}$

m为离子质量（kg），q=ze为电荷（C），z为电荷数，e为元电荷；U为加速电压（V），v为漂移速度（m/s），L为飞行距离（m），t为漂移时间（s）。

输出与成立条件

同一加速电压和路径下，m/q越大，飞行越慢。记录到达时间，再经校准得到m/z；谱横轴为m/z，纵轴为信号强度。

理想式忽略初始速度与加速区长度；实际还需时间零点和质量校准。本文未报告这些仪器参数。

教学推导，非原文公式。基础关系见Shimadzu，MALDI Theory and Basics，II. TOF MS Analysis，公式(1)–(3)。

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TOF演算：质量增至4倍，时间增至2倍

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原文截图 · PDF p.3，Materials and Methods：MALDI靶点与知识库

假设教学数值，非论文测量

设 U=20,000 V，L=1.00 m，z=1；e=1.602×10^{-19} C；1 Da≈1.661×10^{-27} kg。

m=4000 Da≈6.644×10^{-24} kg。

$t = \sqrt{[6.644\times10^{-24} / (6.408\times10^{-15})]}$

$\approx 3.22\times10^{-5}$ s = 32.2 µs

同条件下的比较

1000 Da单电荷离子的计算时间约16.1 µs。质量比4000/1000=4，时间比$\sqrt{4}$=2。

4000 Da、z=1对应m/z数值约4000；若z=2，相同质量的m/z减半，时间缩短为原来的1/$\sqrt{2}$。

本例解释测量映射，不代表VITEK MS的实际电压、路径或分辨率。

教学算例由理想TOF关系计算；本文未给出加速电压、路径和原始飞行时间。

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知识库匹配：VITEK MS的加权分箱分类

The PBC VITEK MITUBE identification workflow also used VITEK MS or MS PRIME for identification, but the sample was prepared from the PBC instead of from an isolated colony. Initially, VITEK MS KB 3.2 was used, but the database was updated to KB 3.3 during the study. All spectra were reanalyzed using KB 3.3, and only data from KB 3.3 are included in this report; isolates initially identified using KB 3.2 but not identified or interpretable with the reference method using KB 3.3 were excluded from this analysis. Isolates were also excluded if the available spectra acquired with KB 3.2 did not reflect the repeat testing procedure when needed with KB 3.3.

原文截图 · PDF p.5，Materials and Methods：KB3.3完整段落

公开的通用系统说明

厂商说明中，谱经背景与噪声处理、峰检测后分入预定质量区间。各区间的峰存在情况与物种特异权重结合，产生物种相对评分，再转为置信表示。

这是加权特征分类，不是把未知原始谱逐张与库存原始谱做最近邻比较。

本文实际报告

最终分析采用KB 3.3，并限制于Box 1所列适用分类群。论文未公开分箱权重、具体阈值或逐样本置信分数。

不能把厂商网页的通用系统描述当作本研究对KB 3.3算法内部参数的验证。

系统背景：bioMérieux，VITEK MS Healthcare，Knowledge Base / Advanced Spectra Classifier。https://www.biomerieux.com/us/en/our-offer/clinical-products/vitek-ms.html

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加权分箱教学算例：评分与拒识

The PBC VITEK MITUBE identification workflow also used VITEK MS or MS PRIME for identification, but the sample was prepared from the PBC instead of from an isolated colony. Initially, VITEK MS KB 3.2 was used, but the database was updated to KB 3.3 during the study. All spectra were reanalyzed using KB 3.3, and only data from KB 3.3 are included in this report; isolates initially identified using KB 3.2 but not identified or interpretable with the reference method using KB 3.3 were excluded from this analysis. Isolates were also excluded if the available spectra acquired with KB 3.2 did not reflect the repeat testing procedure when needed with KB 3.3.

原文截图 · PDF p.5，Materials and Methods：KB3.3完整段落

假设的简化分类器

三分箱输入 x=(1,0,1)，其中1表示检测到峰。候选A权重 wA=(2,−1,1)，候选B权重 wB=(−1,2,0)。

$S_k = \Sigma_i\, w_{ki} x_i$

SA=2×1−1×0+1×1=3；SB=−1。

i为分箱索引（1–3）；k为候选（A或B）；x_i为峰存在指示（0或1）。

假设的判定规则

仅为演示：最高分≥2，且与次高分差≥2才输出。此例A分数3、分差4，故输出A。若污染或弱信号令x=(0,0,1)，最高分仅1，应拒识。

这些权重、阈值和线性式都不是VITEK专有算法，也不是论文数据。评分不自动等于真实正确概率。

w_{ki}是候选k在分箱i的假设权重；S_k是无量纲教学评分。

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鉴定结果、No ID与重复采集

If both spots were considered “No ID” (no identification) by MALDI, then the acquisition was repeated (the laser was “refired”) on both spots. If still receiving a No ID, then the initial suspension was reprocessed by washing; 100 µL was removed from the initial saline tube and placed into a 2 mL microcentrifuge tube. A volume of 0.9 mL of sterile

原文截图 · PDF p.3，Materials and Methods：No ID 后再采集

输出类型

菌种名称：有分类输出；No ID：未能给出鉴定；低区分度：多个候选难以区分。论文的临床最终结果按正确、误鉴定、未鉴定汇总。

置信指标属于分类器输出；实际准确性必须与独立参照比较。高置信不等于绝不会误鉴定。

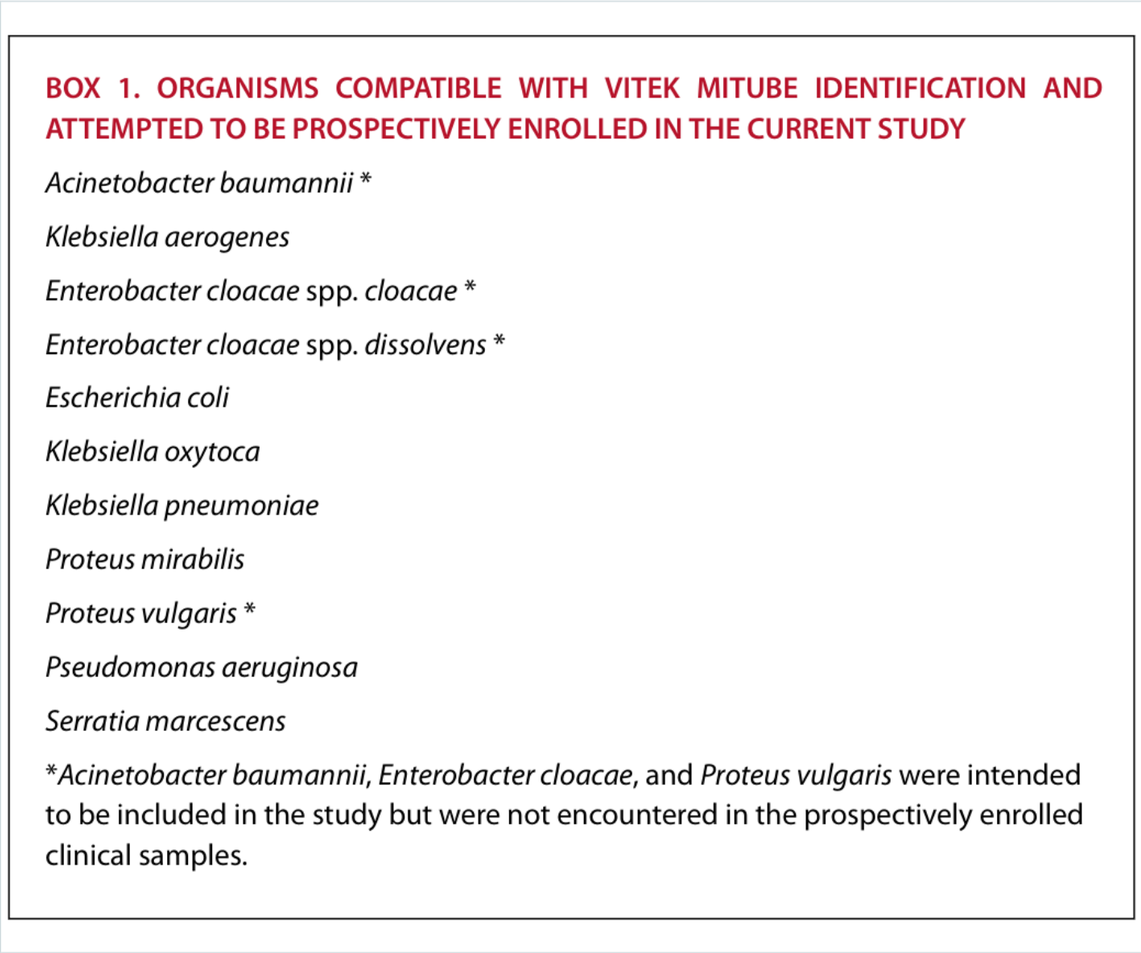
论文的恢复路径

两个原始靶点均No ID时，在原靶点再激发采谱；仍失败时，再处理悬液并制作新靶点。

再激发增加2例成功；洗涤后新靶点没有救回剩余2例。技术重复并不增加独立临床标本数。

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Box 1：本文报告的适用分类群



BOX 1. ORGANISMS COMPATIBLE WITH VITEK MITUBE IDENTIFICATION AND ATTEMPTED TO BE PROSPECTIVELY ENROLLED IN THE CURRENT STUDY

Acinetobacter baumannii *

Klebsiella aerogenes

Enterobacter cloacae spp. *cloacae* *

Enterobacter cloacae spp. *dissolvens* *

Escherichia coli

Klebsiella oxytoca

Klebsiella pneumoniae

Proteus mirabilis

Proteus vulgaris *

Pseudomonas aeruginosa

Serratia marcescens

**Acinetobacter baumannii*, *Enterobacter cloacae*, and *Proteus vulgaris* were intended to be included in the study but were not encountered in the prospectively enrolled clinical samples.

原文截图 · PDF p.3，Box 1 完整名单及脚注

清单与范围

Box 1列出11个名称条目，其中Enterobacter cloacae分成两个亚种；论文按10个物种描述适用范围。

这些名称限定本研究的鉴定任务，并不代表系统可处理任意血培养阳性微生物。

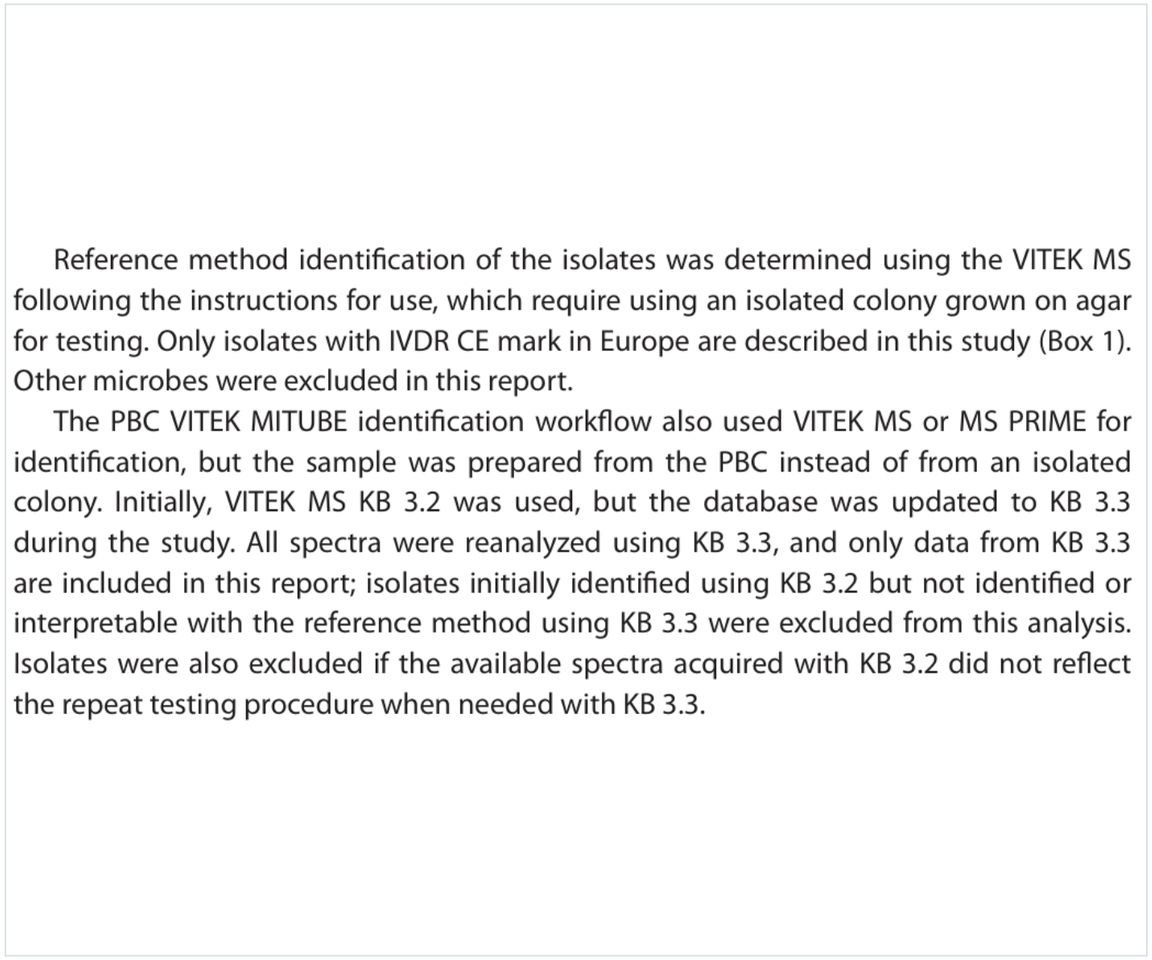
临床未遇到的目标

A. baumannii、E. cloacae和P. vulgaris未在前瞻性临床组遇到，依赖构造样本补充评价。

本文不支持将结果直接推广到革兰阳性菌、真菌或多微生物培养。

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研究设计与参照：比较样本前处理路径



Reference method identification of the isolates was determined using the VITEK MS following the instructions for use, which require using an isolated colony grown on agar for testing. Only isolates with IVDR CE mark in Europe are described in this study (Box 1). Other microbes were excluded in this report.

The PBC VITEK MITUBE identification workflow also used VITEK MS or MS PRIME for identification, but the sample was prepared from the PBC instead of from an isolated colony. Initially, VITEK MS KB 3.2 was used, but the database was updated to KB 3.3 during the study. All spectra were reanalyzed using KB 3.3, and only data from KB 3.3 are included in this report; isolates initially identified using KB 3.2 but not identified or interpretable with the reference method using KB 3.3 were excluded from this analysis. Isolates were also excluded if the available spectra acquired with KB 3.2 did not reflect the repeat testing procedure when needed with KB 3.3.

原文截图 · PDF p.5，Materials and Methods：参照方法与KB3.3

两类验证材料

三个美国临床实验室提供前瞻性剩余阳性培养液；构造样本使用不同分离株补足少见或未遇到的分类群。研究未收集年龄、性别及住院状态。

参照方法

参照为平板分离菌落的VITEK MS鉴定；待测路径由PBC经MITUBE后使用VITEK MS或MS PRIME。

两条路径共享质谱知识库，存在共同分类偏差的可能。仅对那一次不一致，论文报告测序确认P. mirabilis。

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入组、质控与数据库版本的影响

Prospective clinical samples and contrived samples were used for the study. Three clinical sites obtained local institutional review board approval before enrolling clinical samples into the study. Subject-level data, including gender/sex, age, and hospitalization status, were not collected. Remnant PBC from BACT/ALERT 3D (bioMerieux) or BACT/ALERT VIRTUO (bioMerieux) was used in the study, and Fig. 1 depicts the sample workflow for the clinical samples.

For contrived blood culture samples, 10 mL of anticoagulated (sodium polyanethole sulfonate) whole blood was added to a bottle in addition to approximately 50 colony-forming units of bacteria. Different isolates of the same species were used for contrived testing, so the same isolate was not tested repeatedly. Bottles were then incubated on BACT/ALERT 3D or BACT/ALERT VIRTUO.

Blood culture bottles used in the study included BACT/ALERT FA Plus, BACT/ALERT FN Plus, BACT/ALERT PF Plus, BACT/ALERT SA, and BACT/ALERT SN. Positive BACT/ALERT bottles were intended to be tested promptly using the VITEK MITUBE to MALDI workflow and were not tested if more than 16 hours elapsed after bottle positivity was determined by BACT/ALERT 3D or VIRTUO. Before MITUBE processing, a Gram stain and subculture were performed. Samples with mixed morphologies identified by Gram stain were not tested in the study as the MITUBE device is only intended for use with monomicrobial samples.

原文截图 · PDF p.2，Materials and Methods：样本设计与入组

样本与质控限制

选择染色呈单一革兰阴性杆菌的培养液。报阳后超过16小时不测试；≤16小时是入组资格窗，不是检测周转时间。混合形态、当日质控不合格或记录不足等不进入分析；构造样本关联阴性血瓶若污染也排除。

版本统一不等于无选择偏差

研究中KB3.2更新为KB3.3，最终谱统一按KB3.3重分析。缺少必要重复谱或参照结果不能解释的样本被排除。

论文无训练/测试划分：这是固定商业分类器的评价，不是训练新模型。没有独立阴性患者组。

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147例候选如何成为125例分析集

One hundred forty-seven (147) gram-negative prospective samples were considered for the study. Of these 147, 11 were excluded due to low biomass. Of the remaining 136, only 129 were taxa that can be identified by the test system (Box 1). One hundred twenty-five (125) of these 129 were included in the final data set because 4 isolates were excluded. These four isolates were excluded because the available spectra did not allow complete reanalysis with KB3.3, i.e., either there were no available spectra for the repeat testing procedure as it was not necessary with KB3.2 (one isolate) or the reference method results were no longer interpretable with KB3.3 (no identification results for two isolates and low discrimination results for one isolate). These four isolates include one *E. coli* isolate and three *Enterobacter cloacae/asburiae* isolates. Three of these isolates were accurately identified using MITUBE with KB 3.2, and one of the *Enterobacter* species did not achieve identification using the MITUBE workflow and KB 3.2. None of these four isolates were misidentified using KB 3.2. Samples that deviated from the protocol (e.g., samples that were determined to be polymicrobial, isolates with no definitive reference identification, samples processed outside of the timeframes described in the Materials and Methods, and samples with incomplete testing or incomplete documentation of testing) were excluded and not considered in the analysis.

原文截图 · PDF p.5，Results：147至125的筛选

原文给出的筛选算术

147 − 11低生物量 = 136

136 − 7非适用分类群 = 129

129 − 4重分析/参照问题 = 125

4例为1例E. coli和3例E. cloacae/asburiae；其中1例缺少KB3.3所需重复谱，另3例参照结果不再可解释。

123/125描述筛选后的成功比例，不能作为全部147例候选的鉴定率。原文另列方案偏离排除，未提供其完整逐项人数。

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Table 1：前瞻性临床组完整结果

TABLE 1 Performance of the VITEK MITUBE identification compared to the reference standard identification for prospectively enrolled clinical samples* of positive blood culture broth

Organism identification using reference method[a]	Quantity of isolates	Organism identification using the VITEK MITUBE system		
		No identification obtained	Incorrect identification obtained	Accurate species-level identification obtained
Escherichia coli	69	–[b]	–	69 (100%)
Klebsiella aerogenes	4	–	–	4 (100%)
Klebsiella oxytoca	3	–	–	3 (100%)
Klebsiella pneumoniae	29	2 (7%)	–	27 (93%)
Proteus mirabilis	9	–	–	9 (100%)
Pseudomonas aeruginosa	10	–	–	10 (100%)
Serratia marcescens	1	–	–	1 (100%)
Total	125	2 (1.6%)	0 (0%)	123 (98.4%)

[a]*Acinetobacter baumannii*, *Enterobacter cloacae*, and *Proteus vulgaris* were not encountered in the prospectively enrolled clinical samples.
[b]"–" indicates none.

原文截图 · PDF p.5，Table 1 完整表及脚注

最终125例来自7个物种：123例正确、2例未鉴定、0例误鉴定。失败均为K. pneumoniae；E. coli占69/125，物种分布明显不均。

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Table 4：首次采集与再激发分开报告

TABLE 4 Performance of the MITUBE to MALDI workflow in achieving identification of on-label[a] gram-negative bacteria in 125 prospective positive blood culture clinical samples across 3 laboratory sites

Identified with initial attempt	Identified after refiring the laser	Not identified	Misidentified
121 (96.8%)	2 (1.6%)	2 (1.6%)	0 (0%)

[a]See Box 1 for on-label taxa.

原文截图 · PDF p.7，Table 4 完整表及脚注

首次与最终结果

首次121/125=96.8%；再激发额外识别2/125=1.6%，最终123/125=98.4%。仍未识别2/125=1.6%。

“98.4%”包括恢复尝试，不能标为首次成功率。

识别率与准确性

首次未获鉴定的4例中，再激发救回2例，条件恢复比例为2/4=50%（据表计算）。

临床组有输出者123例均与参照相符，123/123=100%；这是条件一致比例，不是对所有目标或未来样本“100%准确”的保证。

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Table 2：合并组完整结果

TABLE 2 Performance of the VITEK MITUBE identification compared to the reference standard identification for prospectively enrolled clinical samples of positive blood culture broth and contrived blood culture broth samples

Organism identification using reference method	Quantity of isolates	Organism identification using the VITEK MITUBE system		
		No identification obtained	Incorrect identification obtained	Accurate species-level identification obtained
Acinetobacter baumannii[a]	25	–[e]	–	25 (100%)
Enterobacter cloacae ssp. *cloacae*[a]	25	–	–	43 (100%)[b]
Enterobacter cloacae ssp. *dissolvens*[a]	2	–	–	2 (100%)
Escherichia coli	75	–	–	75 (100%)
Klebsiella aerogenes	28	–	–	28 (100%)
Klebsiella oxytoca	28	–	–	28 (100%)
Klebsiella pneumoniae	38	2 (5.3%)	–	36 (94.7%)
Proteus mirabilis	31	1 (3.2%)	1 (3.2%)[c]	29 (93.5%)
Proteus vulgaris[a]	31	13 (41.9%)	–	18 (58.1%)
Pseudomonas aeruginosa	33	–	–	33 (100%)
Serratia marcescens	26	–	–	26 (100%)
Total[d]	342	16 (4.4%)	1 (0.3%)	343 (95.3%)

[a]*Acinetobacter baumannii*, *Enterobacter cloacae*, and *Proteus vulgaris* were not encountered in the prospectively enrolled clinical samples.
[b]Eighteen (18) of the *Enterobacter cloacae* ssp. *cloacae* isolates were tested on both VITEK MS and VITEK MS PRIME and thus have two test results for one isolate.
[c]The incorrect identification was obtained when analyzed by KB 3.3. The VITEK MITUBE workflow provided *Enterococcus faecalis,* whereas the isolated colony workflow provided *Proteus mirabilis*. Sequencing identification confirmed *P. mirabilis*.
[d]342 specimens were tested leading to 360 test results[b]. Among them, 125 were prospective clinical specimens (described in Table 1) and 217 were contrived specimens (described in Table 3).
[e]"–" indicates none.

原文截图 · PDF p.6，Table 2 完整表及脚注

表头“Quantity of isolates”共342个标本；结果列合计360次测试。18例Enterobacter cloacae ssp. cloacae在两种质谱平台各测试一次，不能把360称为独立标本数。

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Table 2表体：失败集中在P. vulgaris

TABLE 2 Performance of the VITEK MITUBE identification compared to the reference standard identification for prospectively enrolled clinical samples of positive blood culture broth and contrived blood culture broth samples

Organism identification using reference method	Quantity of isolates	Organism identification using the VITEK MITUBE system		
		No identification obtained	Incorrect identification obtained	Accurate species-level identification obtained
Acinetobacter baumannii[a]	25	–[e]	–	25 (100%)
Enterobacter cloacae ssp. *cloacae*[a]	25	–	–	43 (100%)[b]
Enterobacter cloacae ssp. *dissolvens*[a]	2	–	–	2 (100%)
Escherichia coli	75	–	–	75 (100%)
Klebsiella aerogenes	28	–	–	28 (100%)
Klebsiella oxytoca	28	–	–	28 (100%)
Klebsiella pneumoniae	38	2 (5.3%)	–	36 (94.7%)
Proteus mirabilis	31	1 (3.2%)	1 (3.2%)[c]	29 (93.5%)
Proteus vulgaris[a]	31	13 (41.9%)	–	18 (58.1%)
Pseudomonas aeruginosa	33	–	–	33 (100%)
Serratia marcescens	26	–	–	26 (100%)
Total[d]	342	16 (4.4%)	1 (0.3%)	343 (95.3%)

原文截图 · PDF p.6，Table 2 表体放大（脚注另页放大）

P. vulgaris正确18/31=58.1%，未鉴定13/31=41.9%；它占全部16次未鉴定中的13次（81.25%）。其余未鉴定为K. pneumoniae 2次和P. mirabilis 1次。

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Table 2脚注：343/360与325/342

[a]*Acinetobacter baumannii*, *Enterobacter cloacae*, and *Proteus vulgaris* were not encountered in the prospectively enrolled clinical samples.
[b]Eighteen (18) of the *Enterobacter cloacae* ssp. *cloacae* isolates were tested on both VITEK MS and VITEK MS PRIME and thus have two test results for one isolate.
[c]The incorrect identification was obtained when analyzed by KB 3.3. The VITEK MITUBE workflow provided *Enterococcus faecalis,* whereas the isolated colony workflow provided *Proteus mirabilis*. Sequencing identification confirmed *P. mirabilis*.
[d]342 specimens were tested leading to 360 test results[b]. Among them, 125 were prospective clinical specimens (described in Table 1) and 217 were contrived specimens (described in Table 3).
[e]"–" indicates none.

原文截图 · PDF p.6，Table 2 完整脚注a–e

测试级与标本级

125临床标本 + 217构造标本 = 342。18个额外平台测试使测试总数为360。

343/360=95.28%正确测试；原文同时报告325/342=95.03%正确标本。不要将343直接除以342。

输出率与有输出时的一致比例

有输出 = 343正确 + 1误鉴定 = 344。输出率344/360=95.56%；有输出时正确比例343/344=99.71%。

16/360=4.44%未鉴定，1/360=0.28%误鉴定。以上推算均保留原始分母，不替代原文的95.3%正确鉴定率。

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Table 3：构造样本完整结果

TABLE 3 Performance of the VITEK MITUBE identification compared to the reference standard identification for contrived blood culture broth samples

Organism identification using reference method	Quantity of isolates	Organism identification using the VITEK MITUBE system		
		No identification obtained	Incorrect identification obtained	Accurate species-level identification obtained
Acinetobacter baumannii	25	–[d]	–	25
Enterobacter cloacae ssp. *cloacae*	25	–	–	43[a]
Enterobacter cloacae ssp. *dissolvens*	2	–	–	2
Escherichia coli	6	–	–	6
Klebsiella aerogenes	24	–	–	24
Klebsiella oxytoca	25	–	–	25
Klebsiella pneumoniae	9	–	–	9
Proteus mirabilis	22	1 (4.5%)	1 (4.5%)[b]	20 (90.9%)
Proteus vulgaris	31	13 (41.9%)	–	18 (58.1%)
Pseudomonas aeruginosa	23	–	–	23
Serratia marcescens	25	–	–	25
Total[c]	217	14 (6.0%)	1 (0.4%)	220 (93.6%)

[a]Eighteen (18) of the *Enterobacter cloacae* ssp. *cloacae* isolates were tested on both VITEK MS and VITEK MS PRIME and thus have two test results for one isolate.
[b]The incorrect identification was obtained when analyzed by KB 3.3. The VITEK MITUBE workflow provided *Enterococcus faecalis,* whereas the isolated colony workflow provided *Proteus mirabilis*. Sequencing identification confirmed *P. mirabilis*.
[c]217 specimens were tested leading to 235 test results[a].
[d]"–" indicates none.

原文截图 · PDF p.6，Table 3 完整表及脚注

217个构造标本形成235次测试：220正确、14未鉴定、1误鉴定。构造样本补充物种覆盖，不能消除真实临床组对这些少见目标缺乏验证的限制。

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Table 3脚注：独立标本与重复测试

[a]Eighteen (18) of the *Enterobacter cloacae* ssp. *cloacae* isolates were tested on both VITEK MS and VITEK MS PRIME and thus have two test results for one isolate.
[b]The incorrect identification was obtained when analyzed by KB 3.3. The VITEK MITUBE workflow provided *Enterococcus faecalis,* whereas the isolated colony workflow provided *Proteus mirabilis*. Sequencing identification confirmed *P. mirabilis*.
[c]217 specimens were tested leading to 235 test results[a].
[d]"–" indicates none.

原文截图 · PDF p.6，Table 3 完整脚注a–d

技术重复的分母

同一物种的构造测试使用不同分离株，而18个E. cloacae ssp. cloacae分离株在两平台各测试一次。因此217标本产生235测试。

测试级正确比例220/235=93.62%；不能写成220/217。

验证单位与外推

双平台结果不是18个新增独立标本；把这些结果当作独立样本可能夸大信息量。

构造样本帮助补足分类群，但不复制临床标本所有宿主背景与用药条件。P. vulgaris的31例全部来自构造组。

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Table 3表体与一次跨革兰分类误鉴定

TABLE 3 Performance of the VITEK MITUBE identification compared to the reference standard identification for contrived blood culture broth samples

Organism identification using reference method	Quantity of isolates	Organism identification using the VITEK MITUBE system		
		No identification obtained	Incorrect identification obtained	Accurate species-level identification obtained
Acinetobacter baumannii	25	–[d]	–	25
Enterobacter cloacae ssp. *cloacae*	25	–	–	43[a]
Enterobacter cloacae ssp. *dissolvens*	2	–	–	2
Escherichia coli	6	–	–	6
Klebsiella aerogenes	24	–	–	24
Klebsiella oxytoca	25	–	–	25
Klebsiella pneumoniae	9	–	–	9
Proteus mirabilis	22	1 (4.5%)	1 (4.5%)[b]	20 (90.9%)
Proteus vulgaris	31	13 (41.9%)	–	18 (58.1%)
Pseudomonas aeruginosa	23	–	–	23
Serratia marcescens	25	–	–	25
Total[c]	217	14 (6.0%)	1 (0.4%)	220 (93.6%)

原文截图 · PDF p.6，Table 3 表体放大（脚注见完整表页）

构造P. mirabilis在KB3.3下被MITUBE路径报为E. faecalis；菌落路径报P. mirabilis，测序确认后者。染色筛选与最终分类可能冲突，误鉴定应保留为独立风险。

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结果支持的结论与未覆盖的问题

3. Accurate species-level identification was obtained in 95% (343/360) of the tests, which included accurate identifications for 95% (325/342) of the isolates. No identification was obtained in 4.4% (16/360) of the tests, and most of these (13/16) were due to *P. vulgaris*, which was only successfully identified in about half (18/31) of the attempts. One misidentification occurred in which *P. mirabilis* was misidentified as *E. faecalis*.

Table 4 describes the isolates that were acquired with initial testing and acquired after attempted reacquisition of spectra by refiring on the same target spots. No identifications were obtained by washing the suspension and attempting acquisition from newly created target spots.

原文截图 · PDF p.6，Results：合并结果与失败

可引用的研究结论

在限定的阳性、单微生物、目标革兰阴性菌培养液中，MITUBE结合商业质谱可在不等待分离菌落的路径上获得较高的正确鉴定比例。

不能从本文补出的指标

没有分析检出限、全流程耗时分布、成本核算、临床治疗影响或患者结局；没有逐物种置信区间或多菌混合检出能力验证。

13例P. vulgaris失败说明总体比例不能代替逐物种评价。低生物量排除会限制向困难标本推广。

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在introduction中放置这项技术

Over the past decade, many clinical microbiology laboratories have implemented direct-from-PBC molecular testing to rapidly identify the organism present in the PBC and, in some cases, to characterize the genotypic antimicrobial resistance determinants (5, 6). In the coming decade, we anticipate clinical microbiology laboratories will more widely adopt direct from PBC systems for rapid antibiotic susceptibility testing (AST) as more commercial AST systems achieve regulatory clearance for directly testing PBC. These rapid AST systems will likely be paired with rapid identification systems, and it is not yet clear if clinical laboratories will continue to use molecular panels for rapid identification from PBC or if labs will transition to fast MALDI solutions for identification, such as SepsiTyper (Bruker) or VITEK MITUBE, or to other methods, such as Accelerate Arc. We anticipate the decision to maintain molecular vs transition to MALDI for PBC identification will be influenced by multiple variables, including the rapidity, affordability, simplicity, detection of resistance genes, and accuracy of the MALDI solutions as well as the perceived need to implement rapid AST, which would need to be paired with a rapid identification system. The VITEK MITUBE system has a benefit in that the system could be employed with minimal capital expenditure and little or no new equipment acquisition for laboratories that currently use VITEK MS or MS PRIME.

The VITEK MITUBE system is not marketed in the U.S., and its IVDR CE mark in Europe is limited to frequently encountered gram-negative bacteria (Box 1). Expanding the indications for use to include more taxa could enhance the utility of the system.

原文截图 · PDF p.7，Discussion：应用位置与适用边界

方法位置

本研究属于“阳性血培养后的微生物清理＋蛋白质谱特征分类”。其贡献在样本接口及工作流验证，不是新型质谱分析器，也不是从原血直接检出病原体。

与其他检测路线的关系

论文讨论其与快速药敏联用的可能性，并将鉴定速度、成本、易用性及耐药基因信息作为实验室选择方法时的考虑因素。

这些是作者讨论，不是本研究的头对头经济学或临床效益结果。菌种鉴定不直接给出耐药基因或药敏表型。

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对设备研究的启发与待验证假设

The VITEK MITUBE to MALDI workflow produced accurate identifications of bacteria directly from PBC without the need to subculture to solid media or use a molecular identification system. Initial MALDI acquisition identified 96.8% (121/125) of prospective isolates, and repeat firing on the same target spot increased the identification frequency to 98.4% (123/125). Washing the bacterial suspension and making new target spots did not yield identifications for the two isolates that were not initially identified.

Some of the taxa intended to be included in the study were not encountered in the prospective clinical PBC samples (Table 1), and the taxa encountered infrequently or not at all during prospective enrollment were assessed using contrived samples, which demonstrated similar performance with the exception of *P. vulgaris*, which frequently was not able to be identified using the VITEK MITUBE to MALDI workflow (Table 2) (4).

原文截图 · PDF p.7，Discussion：再激发与P. vulgaris

由本研究可提炼的设计问题

样本前处理决定能否进入传感器；低生物量门槛、类别覆盖和拒识路径都应在系统评价中单独记录。

光学设备若使用培养液作为输入，也应报告背景抑制与样本入选条件，而非只比较分类后准确率。

待验证，不是本文结论

可研究“分离回收率与信号质量是否相关”，或“弱信号触发重复采集能否减少拒识”。必须新增配对数据和事先定义的指标。

本论文未给出质谱原始数据，因此不能据此比较光学与质谱的分辨率、LOD或泛化性能。

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来源、范围与利益关系



This study was sponsored by bioMérieux. D.D.R. has performed, is performing, or has engaged in the potential for funding of sponsored research projects funded through his employers with the following entities: Abbott, Altona, BD, bioMérieux, Cepheid, Cleveland Diagnostics, HelixBind, Hologic, Luminex, Pattern Bio, Qiagen, Q-Linea, Rapid Diagnostics, Roche, Selux Diagnostics, Specific Diagnostics, Thermo Fisher, and Vela Diagnostics; D.D.R. has equity and/or receives scientific advisory compensation from Renascent Diagnostics and Next Gen Diagnostics. N.D.H. has engaged in research from investigator initiated grants from Merck and the FDA and funded projects from bioMérieux. K.R. and J.G. are bioMérieux employees who coordinated the study at sponsor level. Y.X.Y. has no disclosures. D.J.H. has performed and is performing sponsored research projects funded by bioMerieux, INC and bioMerieux France.

原文截图 · PDF p.1，资助与利益关系

本文来源与版本

主文8页，Fig.1–2、Table 1–4及Box 1。两图无字母子图；PMC XML及出版商页面均未列出补充材料。未见更正或撤稿提示。

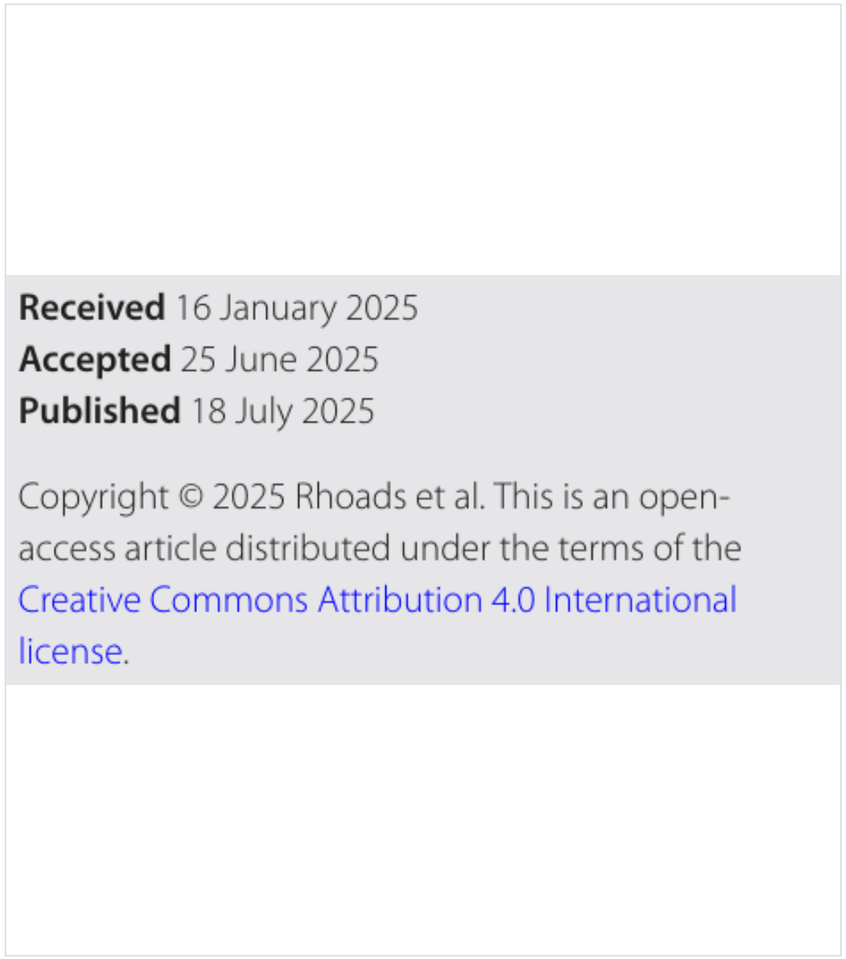
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利益关系

bioMérieux资助；两位作者是其雇员。设备评价结果需结合研究设计与外部验证判断。

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原理与制作方法来源



Received 16 January 2025
Accepted 25 June 2025
Published 18 July 2025

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技术背景，不替代本文数据

Shimadzu：MALDI Theory and Basics，MALDI与TOF基础。

bioMérieux：VITEK MS Healthcare，Advanced Spectra Classifier与加权分箱说明。

制作方法记录

已使用并阅读no-ai-slop与Scientific Writing 2.3的实际文件及证据、写作、图表参考。

方法文献：Kassis等，Scientific Agent Skills (2026)，doi:10.48550/arXiv.2609.00065；仅为教学制作流程来源，不是微生物检测证据。

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要点

- 阳性培养液输入
- 分离后测质谱
- 识别率与准确性分开
- 保留拒识和物种差异

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